Mean-Field Pontryagin Maximum Principle

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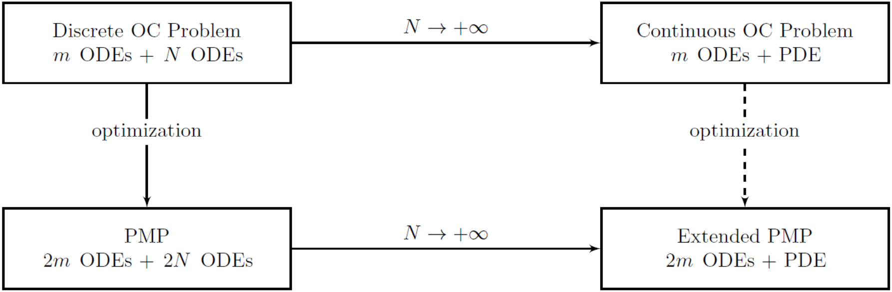
Goal of this paper



Sparse Mean-field Optimal Control Theorem

Problem 1

For $T > 0$ fixed, find $u^* \in L^1([0,T];\mathcal{U})$ minimizing the cost functional

$$F(u) = \int_0^T [L(y(t),\mu(t)) + \gamma(u(t))]dt \tag{1}$$

where (y,μ) solve

$$\begin{cases} \dot{y}_k(t) = (K \star \mu(t))(y_k(t)) + f_k(y(t)) + B_k u(t), \quad k = 1,\cdots,m, \\ \partial\mu(t) = -\nabla_x \cdot [(K \star \mu(t) + g(y(t)))\mu(t)], \end{cases}$$

for the given initial datum $(y(0),\mu(0)) = (y^0,\mu^0) \in \mathbb{R}^{dm} \times \mathcal{P}_c(\mathbb{R}^d)$.

Main Theorem

Theorem (1.1 extended Pontryagin Maximum Principle)

Fix ans initial datum $(y^0, \mu^0) \in \mathbb{R}^{dm} \times \mathcal{P}_c(\mathbb{R}^d)$ *and assume that Hypotheses holds, Then there exists a mean-field optimal control for Problem 1.*
Furthermore, if u^* *is a mena-field optimal control for Problem 1 and* (y^*, μ^*) *is the corresponding trajectory, then* (u^*, y^*, μ^*) *satisfies the following extended Pontryagin Maximum Principle :*

extended Pontryagin Maximum Principle

extended Pontryagin Maximum Principle

There exits $(q^*(\cdot), \nu^*(\cdot)) \in Lip([0.T]; \mathbb{R}^{dm} \times \mathcal{P}_1(\mathbb{R}^{2d}))$ such that

- there exists $R_T > 0$, depending only on $y^0, supp(\mu^0), d, K, g, f_k, B_k, \mathcal{U}, and\ \ T$, such that $supp(\nu^*(\cdot)) \subset B(0, R_T)$ and it satisfies $nu^*(t)(E \times \mathbb{R}^d = \mu^*(t)(E)$ for all $t \in [0, T]$ and for every Borel set $E \subset \mathbb{R}^d$
- it holds

$$\begin{cases} \dot{y}_k^* & = \nabla_{q_k} \mathbb{H}_c(y^*, q^*, \nu^*, u^*), \\ \dot{q}_k^* & = -\nabla_{y_k} \mathbb{H}_c(y^*, q^*, \nu^*, u^*), \\ \partial_t \nu^* & = -\nabla_{(x,r)} \cdot ((J\nabla_\nu \mathbb{H}_c(y^*, q^*, \nu^*, u^*))\nu^*), \\ u^* & = \arg\max_{u \in \mathcal{U}} \mathbb{H}_c(y^*, q^*, \nu^*, u^*) \end{cases} \tag{2}$$

extended Pontryagin Maximum Principle

extended Pontryagin Maximum Principle

- where $J \in \mathbb{R}^{2d \times 2d}$ is the symplectic matrix

$$J = \begin{pmatrix} 0 & Id \\ -Id & 0 \end{pmatrix}$$

the Hamiltonian $\mathbb{H}_C : \mathbb{R}^{2dm} \times \mathcal{P}_c(\mathbb{R}^2 d) \to \mathbb{R}$ is defined as

$$\mathbb{H}_c(y, q, \nu, u) = \begin{cases} \mathbb{H}(y, q, \nu, u) & if \;\; supp(\nu) \subset \overline{B(0, R_T)}, \\ \infty & elsewhere; \end{cases}$$

- $\mathbb{H}$ is defined as

$$\begin{aligned}\mathbb{H}(y,q,\nu,u) =& \frac{1}{2}\int_{\mathbb{R}^{4d}}(r-r')\cdot K(x-x')d\nu(x,r)d\nu(x',r') \\ &+ \int_{\mathbb{R}^{2d}} r\cdot g(y)(x)d\nu(x,r) \\ &+ \sum_{k=1}^{m}\int_{\mathbb{R}^{2d}} q_k\cdot K(y_k-x)d\nu(x,r) \\ &+ \sum_{k=1}^{m} q_k\cdot(f_k(y)+B_k u) \\ &- L(y,\pi_1\nu)-\gamma(u)\end{aligned}$$

- the following conditions for systems 15 hold at time 0 : $y^*(0)=y^0$ and $\nu^*(0)(E\times\mathbb{R}^d)=\mu^0(E)$ for every Borel set $E\subset\mathbb{R}^d$
- the following conditions for systems 15 hold at time T : $q^*(T)=0$ and $\nu^*(T)(\mathbb{R}^d\times E)=\delta_0(E)$ for every Borel set $E\subset\mathbb{R}^d$

Sparse Mean-field Optimal Control Theorem

Problem 1

For $T > 0$ fixed, find $u^* \in L^1([0,T];\mathcal{U})$ minimizing the cost functional

$$F(u) = \int_0^T [L(y(t), \mu(t)) + \gamma(u(t))]dt \tag{3}$$

where (y, μ) solve

$$\begin{cases} \dot{y}_k(t) = (K \star \mu(t))(y_k(t)) + f_k(y(t)) + B_k u(t), \quad k = 1, \cdots, m, \\ \partial \mu(t) = -\nabla_x \cdot [(K \star \mu(t) + g(y(t)))\mu(t)], \end{cases}$$

for the given initial datum $(y(0), \mu(0)) = (y^0, \mu^0) \in \mathbb{R}^{dm} \times \mathcal{P}_c(\mathbb{R}^d)$.

Sparse Optimal Control Theorem

Problem 2

For $T > 0$ fixed, find $u^* \in L^1([0, T]; \mathcal{U})$ minimizing the cost functional

$$F(u) = \int_0^T [L(y(t), \mu(t)) + \gamma(u(t))]dt \tag{4}$$

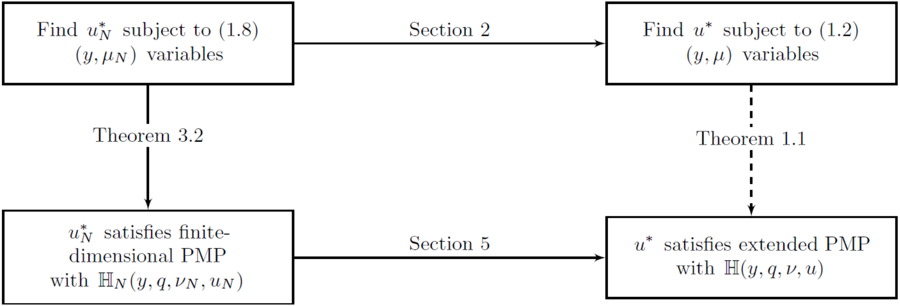
where (y, μ) solve

$$\begin{cases} \dot{y}_k(t) = \frac{1}{N}\sum_{j=1}^N K(y_k - x_j) + f_k(y(t)) + B_k u(t), & k = 1, \cdots, m, \\ \dot{x}_i(t) = \frac{1}{N}\sum_{j=1}^N K(x_i - x_j) + g(y)(x_i), & k = 1, \cdots, N \end{cases}$$

for the given initial datum $(y(0), x(0)) = (y^0, x^0) \in \mathbb{R}^{dm} \times \mathbb{R}^{dN}$, where

$$\mu_N(t)(x) = \frac{1}{N}\sum_{i=1}^N \delta(x - x_i(t)),$$

Goal of this paper



Hypotheses

- The function $K \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R}^d)$ is odd and sublinear, i.ek., there exists $C_K > 0$ such that for all $x \in \mathbb{R}^d$ it holds

$$\|K(x)\| < C_K(1 + \|x\|). \tag{5}$$

- The function $L : \mathbb{R}^{dm} \times \mathcal{P}_1(\mathbb{R}^d \to \mathbb{R}$ is

$$L(y, \mu) = \int_{\mathbb{R}^d} l(y, x, \int w\mu) d\mu(x). \tag{6}$$

  with $l \in \mathcal{C}^2(\mathbb{R}^{dm} \times \mathbb{R}^d \times \mathbb{R}^d; \mathbb{R})$ and $w \in \mathcal{C}^2(\mathbb{R}^d, \mathbb{R}^d)$.

- The function $g \in \mathcal{C}^2(\mathbb{R}^{dm}; \mathcal{C}^2(\mathbb{R}^d; \mathbb{R}^d))$ satisfies for all $x \in \mathbb{R}^d$ and for all $y \in \mathbb{R}^{dm}$

$$g(y)(x) \cdots \leq G_1\|x\|^2 + g_2 \max_{l=1,\cdots,m} \|y\|^2 + G_3 \tag{7}$$

  where the G_1, G_2, G_3 are constants independent on x and y

Hypotheses

- For each $k = 1, \cdots, m$, the function $f_k \in \mathcal{C}^2(\mathbb{R}^{dm}; \mathbb{R}^d)$ satisfies for all $y \in \mathbb{R}^{dm}$
$$f_k(y) \cdot (y_k) \leq F_1 \max_{l=1,\cdots,m} \|y_l\|^2 + F_2. \tag{8}$$
where the constants F_1, F_2 are independent on y and k.
- The set $\mathcal{U} \subset R^D$ is compact and convex
- The function $\gamma : \mathcal{U} \to \mathbb{R}$ is strictly convex.

Cucker-Smale model

CS model

$$\begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^{N} \phi(\|x_i - x_j\|)(v_j - v_i), \quad i = 1, \cdots, N, \end{cases} \tag{9}$$

Mean-field CS Model

$$\begin{cases} \dot{y}_k = w_k \\ \dot{w}_k = (\Phi \star \mu)(y_k, w_k) + \frac{1}{m} \sum_{j=1}^{m} \phi(\|y_k - y_j\|)(w_j - w_k) + u_k, i = 1, \cdots, m \\ \partial_t \mu = -v \cdot \nabla_x \mu - \nabla_v \cdot [(\Phi \star \mu + \frac{1}{m} \sum_{j=1}^{m} \phi(\|x - y_j\|)(w_j - v))\mu] \end{cases} \tag{10}$$

settings

We define leader state variables $Y_k := \begin{pmatrix} y_k \\ w_k \end{pmatrix}$ and the ones for the followers are $X = \begin{pmatrix} x \\ v \end{pmatrix}$

Settings of functions

$$K(X)\begin{pmatrix} 0 \\ \Phi(x) \end{pmatrix}, f_k(Y) = \begin{pmatrix} w_k \\ \frac{1}{m}\sum_{j=1}^{m} \phi(y_k - y_j)(w_j - w_k) \end{pmatrix} \tag{11}$$

$$g(Y)(X) = \begin{pmatrix} v \\ \frac{1}{m}\sum_{j=1}^{m} \phi(x - y_j)(w_j - v) \end{pmatrix} \tag{12}$$

settings

A standard problem in the study of the CS model is to find conditions to ensure flocking, i.e. alginment of the whole crowd to the same velocity. For this reason, it is interesting in our case to study the minimization of the *variance*[2] of the crowd, by choosing

$$L(Y,\mu) := \int_{\mathbb{R}^{2d}} \left(\frac{2}{m} \sum_{k=1}^{m} \|w_k\|^2 + 2\|v\|^2 \right) d\mu(x,v) \tag{13}$$

$$-\|\frac{1}{m}\sum_{k=1}^{m} w_k + \int v d\mu(x,v)\|^2 \tag{14}$$

Hamiltonian of CS Model

For simplicity we will only consider 1-dimensional problem. So we have the following Hamiltonian

- $\mathbb{H}$ is defined as

$$
\begin{aligned}
\mathbb{H}(Y, Q, \nu, u) =& \frac{1}{2} \int_{\mathbb{R}^8} (s - s')\phi(x - x')(v' - v) d\nu(x, v, r, s) d\nu(x', v', r', s \\
&+ \int_{\mathbb{R}^4} (rv + s \frac{1}{m} \sum_{j=1}^{m} \phi(x - y_j)(w_j - v) d\nu(x, v, r, s) \\
&+ \sum_{k=1}^{m} \left(q_k w_k + z_k \int_{\mathbb{R}^4} \phi(y_k - x)(v - w_k) d\nu(x, r) \right) \\
&+ \sum_{k=1}^{m} (z_k \frac{1}{m} \phi(y_k - x)(v - w_k) + z_k u_k) \\
&- L(y, \pi_1 \nu) - \|u\|^2
\end{aligned}
$$

Pontryagin Maximum Principle

Theorem 3.2

Let u_N^* be a solution of Problem 2 and denote $(y^*(\cdot), x^*(\cdot))$ the corresponding trajectory. Then there exists $(q^*(\cdot), \nu^*(\cdot)) \in Lip([0.T]; \mathbb{R}^{dm} \times \mathcal{P}_1(\mathbb{R}^{2d}))$ such that

$$\begin{cases} \dot{y}_k^* & = \nabla_{q_k} \mathbb{H}_c(y^*, q^*, x^*, p^*, u^*), \\ \dot{q}_k^* & = -\nabla_{y_k} \mathbb{H}_c(y^*, q^*, x^*, p^*, u^*), \\ \dot{x}_i^* & = \nabla_{p_i} \mathbb{H}_c(y^*, q^*, x^*, p^*, u^*), \\ \dot{p}_i^* & = -\nabla_{x_i} \mathbb{H}_c(y^*, q^*, x^*, p^*, u^*), \\ u^* & = \arg\max_{u \in \mathcal{U}} \mathbb{H}_c(y^*, q^*, x^*, p^*, u^*) \end{cases} \tag{15}$$

Pontryagin Maximum Principle

Theorem 3.2

Where the Hamiltonian is given by

$$
\begin{aligned}
H_c(y,q,x,p,u) = \sum_{i=1}^{N} p_i \cdot \left(\frac{1}{N} \sum_{j=1}^{N} K(x_i - x_j) + g(y)(x) \right) \qquad (16)\\
+ \sum_{i=1}^{m} q_i \cdot \left(\frac{1}{N} \sum_{j=1}^{N} K(y_i - x_j) + f_i(y) + B_k u \right) \qquad (17)\\
- L(y, \mu_N) - \gamma(u) \qquad (18)
\end{aligned}
$$

Pontryagin Maximum Principle

Using Theorem 3.2 we have the following result for u_k^* in the finite case.

$$u_k^*(Y, Q) = \begin{cases} \frac{1}{2} z_k & if \;\; z_k \in [-2, 2], \\ sign(z_k) & otherwise \end{cases}$$

Pontryagin Maximum Principle

Using Theorem 3.2 we have the following result for u_k^* in the finite case.

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